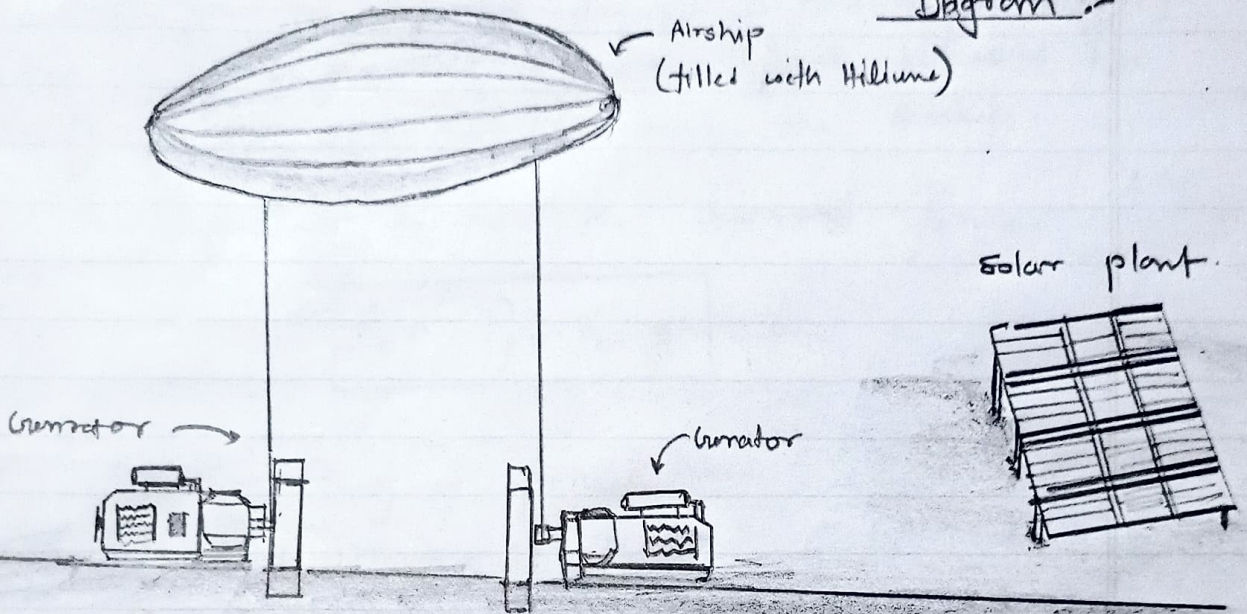


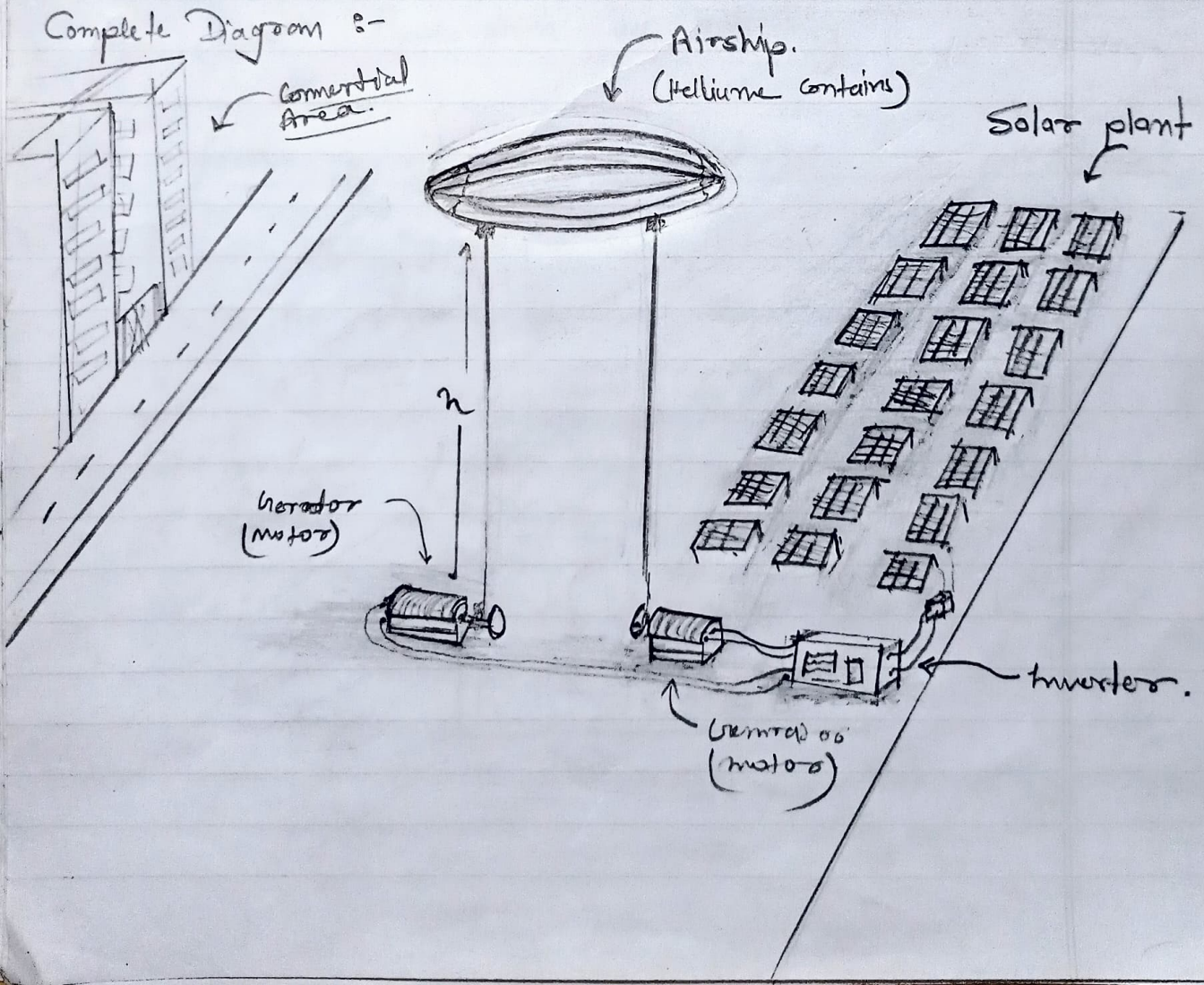
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Airship based energy storage system

The Energy Storage using Airships :-

Overview Diagram :-



Complete Diagram :-



The force generated by the Airship :-

The force generated by the Airships is known as Buoyant force which is generated by the difference between the Density of air and the fill inside the Airship.

formula to find the Buoyant force.

$$F_b = \rho_f \cdot V_b \cdot g$$

Where

$F_b \rightarrow$ Buoyant force

$\rho_f \rightarrow$ Density of fluid.

$V_b \rightarrow$ Volume of the Airship.

$g \rightarrow$ Acceleration due to gravity.

The force generated by an medium size airships :-

$\rho_{air} = 1.225 \text{ Kg/m}^3$ (air density at sea level)

V is the volume of the ship.

$g = 9.8 \text{ /ms}^2$ (acceleration due to gravity)

The volume of an medium size airships is around 15000 m^3

Then from the formula $F_b = \rho_f \cdot V_b \cdot g$.

$$F_b = 18,0375 \text{ N}$$

$\rightarrow$ which is sufficient
to run the motor
of generator.

Torque Required to Run a grid Scale Generator :-

Generally the grid scale generator produce power in term of megawatts So let's take 10 MW generator

The generators generally gives 50 Hz frequency at 3000 RPM

Then the Torque Required is

$$T = \frac{P}{\omega}$$

where

P $\rightarrow$ power output in watts.

T $\rightarrow$ Torque in Newtons meter

ω $\rightarrow$ Angular velocity in radians per second.

$$\omega = \frac{2\pi \cdot \text{RPM}}{60}$$

$$T = \frac{P \cdot 60}{2\pi \cdot \text{RPM}}$$

$$T = \frac{10 \times 10^6 \cdot 60}{2\pi \cdot 3000}$$

$$T = 3183.1 \text{ N/m}$$

where

$$T = F \cdot r$$

if the radius of the rotor is 2m

$$F = 1591.55 \text{ N}$$

This much of force is Required to Rotate The Rotor.

where an medium size Airship produce $153,900 \text{ N}$ of force
It means we can Run $96,10 \text{ MW}$ generators on it

By adding the weight of the Airship and other friction losses we can easily Run $81,10 \text{ MW}$ generators which is huge.

Affect on force due to height :-

Generally the airships do not have any significant affect of height until the do not go to very high.

But it is has some effect of height

when the height increase $\uparrow$

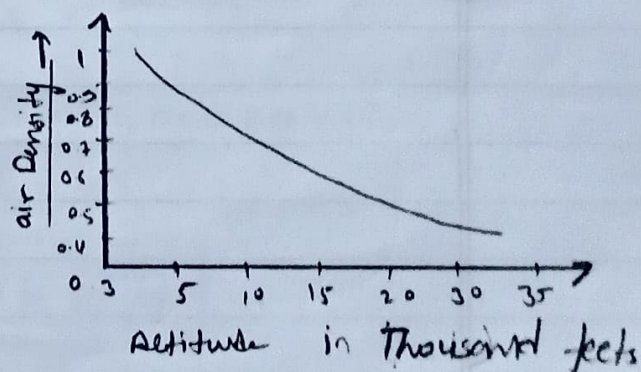
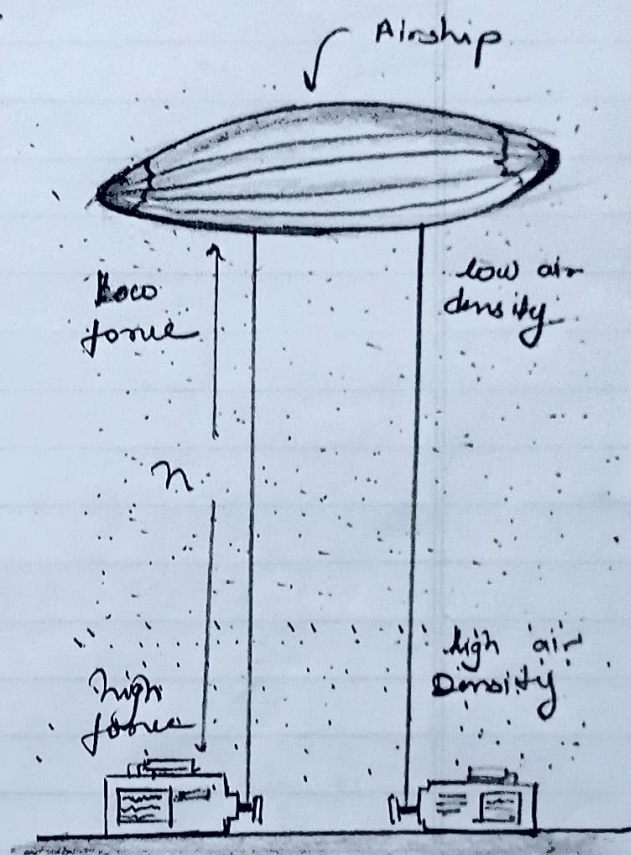
The density of air will decrease $\downarrow$

Then the force will we also decrease $\downarrow$

Maximum height attained by the airships :-

The operation Amplitude of an airship is around 3000 m

The maximum Amplitude of an airship is 7000 m



Power Stored by the airship:-

$$\text{Helium Density} \rightarrow 0.1786 \text{ kg/m}^3$$

$$\text{Air Density} \rightarrow 1.225 \text{ kg/m}^3$$

$$\text{Volume of airship} \rightarrow 1500 \text{ m}^3$$

$$\text{Gravitational Acceleration} \rightarrow 9.8 \text{ m/s}^2$$

$$F_b = (\rho_{\text{air}} - \rho_{\text{helium}}) \cdot g$$

$$F_b = 10.26 \text{ N/m}^3$$

$$F_T = 10.26 \times 15000 \text{ m}^3$$

$$F_T = 153,900 \text{ N}$$

Potential Energy stored if we consider the maximum height is 3000 m.

$$PE = 153,900 \times 3000$$

$$PE = 461,700,000 \text{ J}$$

Electrical energy produced.

Assuming the efficiency of the generator is 85%.

$$E_{\text{electrical}} = PE \times \eta$$

$$E_{\text{electrical}} = 461,700,000 \times 0.85$$

$$E_{\text{electrical}} = 392,445,000 \text{ J}$$

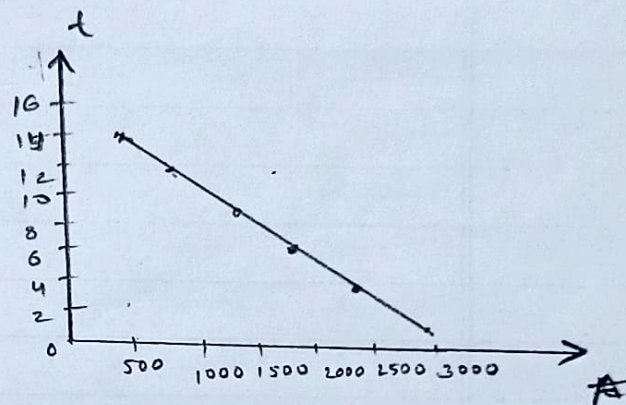
$$\text{Power} = 109 \text{ kW/h}$$

The effects of environment on the Airship:-

There are mainly three environmental factors that affect the Airships. ① Temperature ② pressure. ③ wind.

Temperature:-

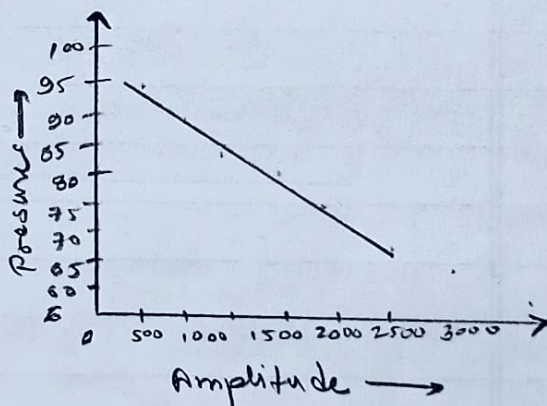
When the temperature decreases the gasses start contracting as a result the density of the gas will increase. This affects our Airship because the temperature decreases with ~~increase~~ in Amplitude.



Temperature-Amplitude graph.

Pressure:-

When the pressure will decrease the gasses start expanding as a result density of the gas starts decreasing it is helpful to our Airship to generate sufficient force.



Pressure-Amplitude graph.

Wind:-

Wind also affects the Airships but now we have the lots of experience to deal with it. That's why the wind is not the major factor.